



DEPARTMENT OF
COMPUTER SCIENCE



PROBLEM SOLVING STRATEGIES

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LEARNING OBJECTIVES

At the end of this lesson, you should be able to:

- Explain what problem solving strategies are.
- Describe the benefits of employing strategies in solving computational problems.
- Apply various problem solving strategies to computational problems.



PROBLEM SOLVING STRATEGIES

- When faced with a problem, we need to thoroughly understand the problem.
- The next step will be how to solve the problem.



PROBLEM SOLVING STRATEGIES

- A problem-solving strategy is a plan used to find a solution or overcome a challenge.
- Each problem-solving strategy includes multiple steps that provides helpful guidelines on how to resolve a business problem or industry challenge.



PROBLEM SOLVING STRATEGIES

- Having a clear understanding of different problem solving strategies is very vital since different problems require different strategies to obtain the best solution.
- So, it is important to consider different strategies with which a problem can be solved.

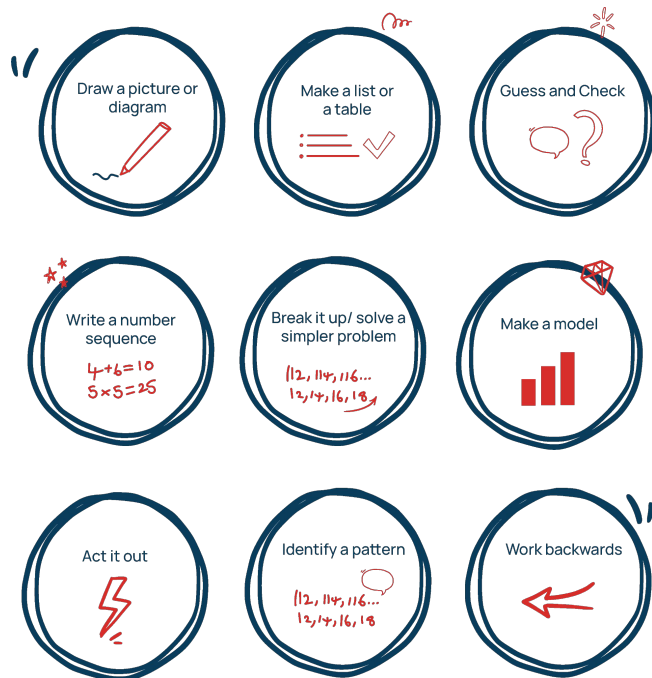


PROBLEM SOLVING STRATEGIES

- After choosing an appropriate strategy, the rules and procedures of that strategy must be studied.
- An efficient strategy will produce a robust and reliable solution to the problem.



PROBLEM SOLVING STRATEGIES





BRAINSTORMING

- Brainstorming is a group focused technique where people come together to solve a problem.
- It involves critical thinking and coming up with various solutions spontaneously.





BRAINSTORMING

- With brainstorming, the ideas of other people can be fine-tuned.
- Avoid criticism of ideas and gather as many ideas as possible.





BRAINSTORMING: EXAMPLE

The image below shows a brainstorming approach.

Topic	Narrower Topic	Even Narrower
Global Warming	Environment	- rising sea levels - destruction of rain forests - air pollution
	Political	- Kyoto Protocol - roles of government
	Human Element	- impact on world health - reducing use of fossil fuel
	Economic	- agriculture - role of corporations
	Geographical	- developing countries - Antarctic region



HEURISTICS

- Helps to obtain solution to problems can be obtained while saving time and energy.
- Decisions are made without stopping to think about the next course of action.



HEURISTICS

- Problems are solved in a fast and efficient manner but solutions are not always rational.
- Relevant to situations where sub-optimal solutions to a problem is acceptable.



HEURISTICS: EXAMPLE

Here are examples of where heuristics can be used to solve problems.

- Rule of thumb
- Choosing between different brands of food items at the grocery store
- Migrating to another country without being familiar with the area



TRIAL AND ERROR

- Deals with recognising and removing errors or failures through various experimental trials.
- Various approaches are implemented until a solution to the problem is arrived at.



TRIAL AND ERROR

- Useful in instances where number of possible solutions to the problem are few
- Time consuming and inefficient in instances where many possible solutions exist



TRIAL AND ERROR: EXAMPLE

How trial and error can be used to solve problems

- Trying out various jobs before choosing a career path
- Living in various locations before concluding on a new home
- Trying out various interests before concluding on a favourite



MID LESSON QUESTIONS

Which of the following is not true about the trial-and-error problem solving strategy?

- a) It can be time consuming when the number of possible solutions to the problem are not few
- b) It is very efficient when the number of possible solutions to the problem are not few**
- c) It involves trying various solutions until the problem is solved
- d) It is recommended for cases where the number of possible solutions to the problem are few



MEANS-ENDS ANALYSIS

- It deals with thinking about the solution to a problem beforehand, determining the best strategy to adopt based on the issues that will be encountered while trying solve the problem.



MEANS-ENDS ANALYSIS

- It involves:
 - breaking down a problem into smaller problems and,
 - solving the smaller problems with the best possible approach until the main problem is solved.



MEANS-ENDS ANALYSIS: EXAMPLE

Areas where a means-ends analysis can be used to solve problems

- Having the target of completing the academic syllabus for a degree in six months.
- A company expects a turnover of 2 billion dollars in a year.



ABSTRACTION

- This is a technique of filtering out the unneeded attributes of the problem to be solved so that we can concentrate on the important parts.



ABSTRACTION

- All aspects or patterns of the problem that will not help us arrive at a solution is removed.
- It helps to form an essential idea of what the problem is and how to solve it.
- The idea formed is also known as a model.



ABSTRACTION: EXAMPLE

The London Underground map presents a good example of abstraction.





ANALOGY

- Deals with using previously existing understanding to understand a new similar system as long as the underlying principles are the same.



ANALOGY

Steps to follow includes:

- Define the problem and generate analogies.
- Find similarities and select any of the analogies.
- Use description and similarities to generate ideas.



ANALOGY: EXAMPLE

We will now discuss some examples where an analogy can be used to solve problems;

- A manager looking for ways to deal with a problematic employee.
- A doctor wanting to treat a patient with a malignant, inoperable tumour buried deep inside the body with an high-intensity ray.



FURTHER READING RESOURCES

- Monica, A. (2017). Brainstorming: Thinking-Problem Solving Strategy. Er. Manoj Kumar. Int. Journal of Engineering Research and Application, 7(3), 33-37.
- Spielman, R. M., Dumper, K., Jenkins, W., Lacombe, A., Lovett, M., & Perlmutter, M. (2021). Problem Solving. Psychology-H5P Edition.
- Knoblock, C. A. (1991). Automatically Generating Abstractions for Problem Solving. CARNEGIE-MELLON UNIV PITTSBURGH PA DEPT OF COMPUTER SCIENCE.
- Hesse, F. W., & Klecha, D. (1990). Use of analogies in problem solving. Computers in Human Behavior, 6(1), 115-129.



REFERENCES

1. *Monica, A. (2017). Brainstorming: Thinking-Problem Solving Strategy. Er. Manoj Kumar. Int. Journal of Engineering Research and Application, 7(3), 33-37.*
2. *Spielman, R. M., Dumper, K., Jenkins, W., Lacombe, A., Lovett, M., & Perlmutter, M. (2021). Problem Solving. Psychology-H5P Edition.*



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THANK YOU